

# NpuKit

## 8×8 systolic array — cycle-by-cycle walkthrough

Same stimulus as sim/systolic\_array\_tb.sv · skewed feed · finish in  $3N-2 = 22$  clocks

Each PE keeps a running  $C[i][j]$  accumulator. A streams west→east; B streams north→south.

**A** ( $A[i][k] = i+1$ )

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |

**B** (all ones)

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

**C = A·B** (expected)

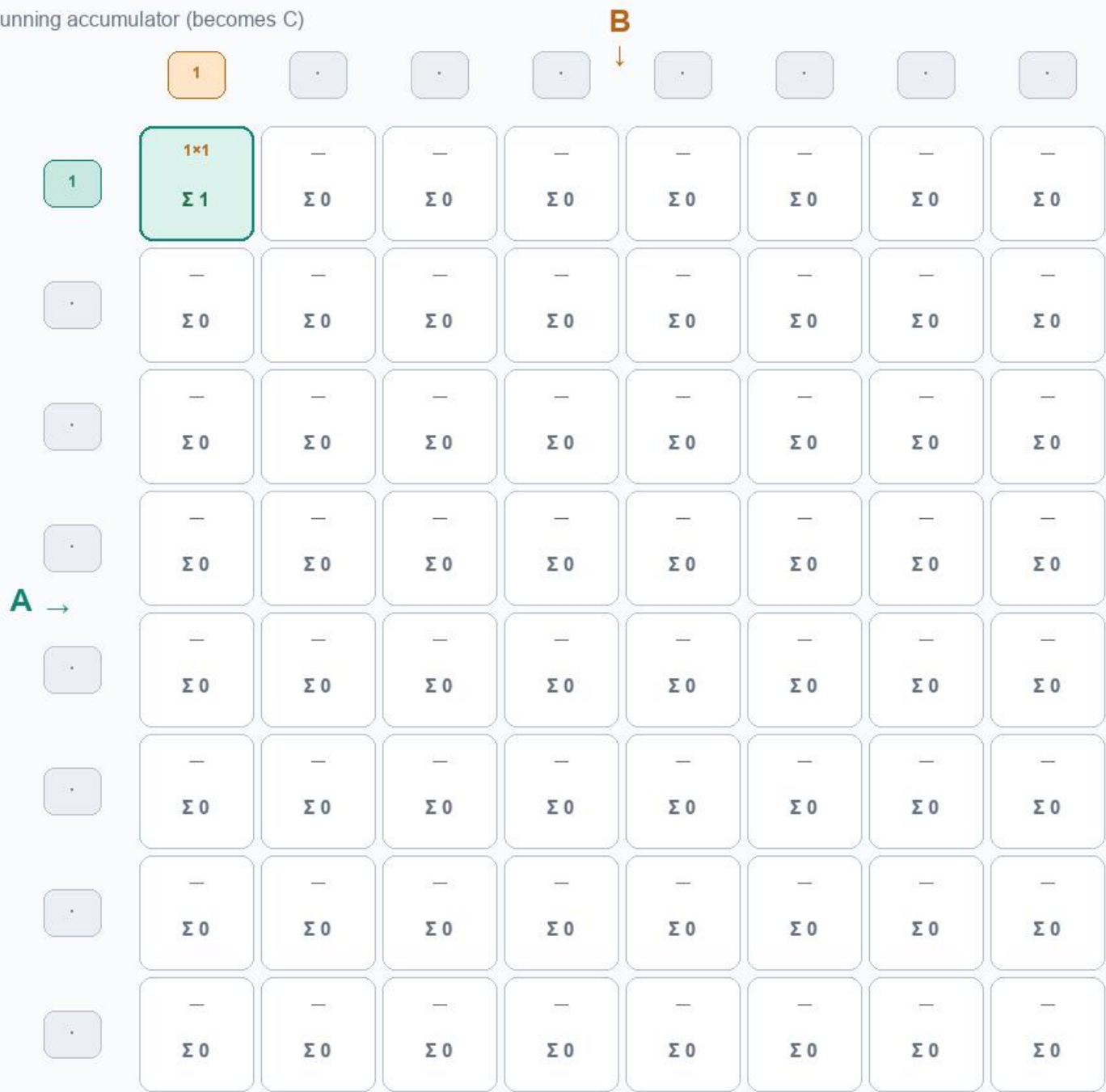
|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 8  | 8  | 8  | 8  | 8  | 8  | 8  | 8  |
| 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 |
| 24 | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| 32 | 32 | 32 | 32 | 32 | 32 | 32 | 32 |
| 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 |
| 48 | 48 | 48 | 48 | 48 | 48 | 48 | 48 |
| 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 |
| 64 | 64 | 64 | 64 | 64 | 64 | 64 | 64 |

### How to read the following pages

- One page per clock cycle ( $t = 0 \dots 21$ ).
- Left pills = A values entering from the west; top pills = B values entering from the north.
- Inside each PE cell:
  - $A \times B$  — operands present this cycle (blank if idle / zeros)
  - $\Sigma$  — running accumulator after this cycle's MAC (this is  $C[i][j]$ )
- Active cells (doing a multiply-accumulate) are tinted mint.
- Final page shows C held in the 64 PE accumulators.

Expected result:  $C[i][j] = 8 \cdot (i+1) \rightarrow$  row 0 = 8, row 1 = 16, ..., row 7 = 64

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

mint cell = MAC this cycle

C stays in PE

**Cycle 2 of 22 (t = 1)**

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)

clock 2/22

streaming MACs



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

```
mint cell = MAC this cycle
```

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell: A×B (operands this cycle)

$\Sigma$  = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE



A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



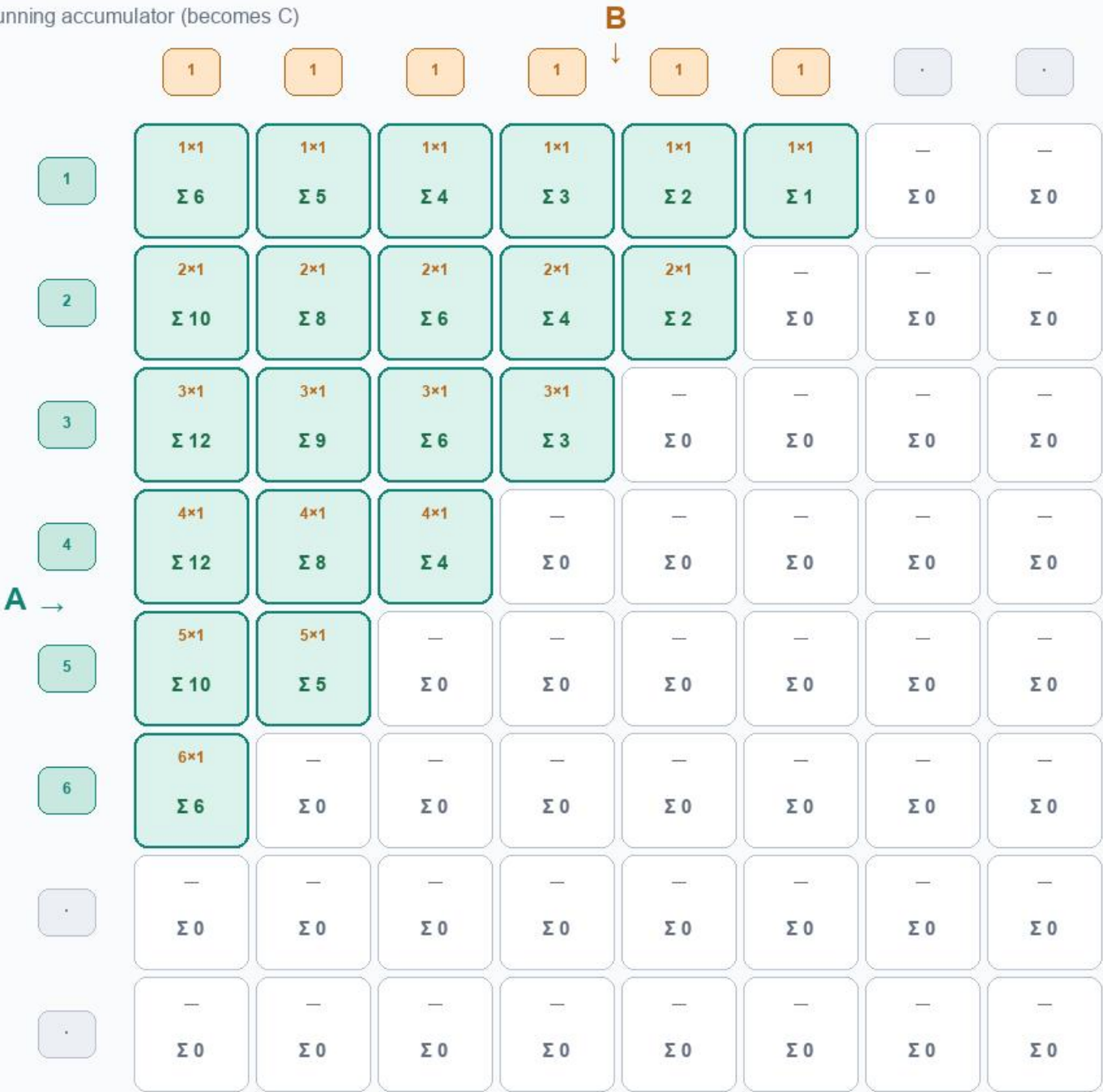
cell: AxB (operands this cycle)

$\Sigma$  = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE · Σ is the running accumulator (becomes C)



cell: A×B (operands this cycle)

Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell: A×B (operands this cycle)

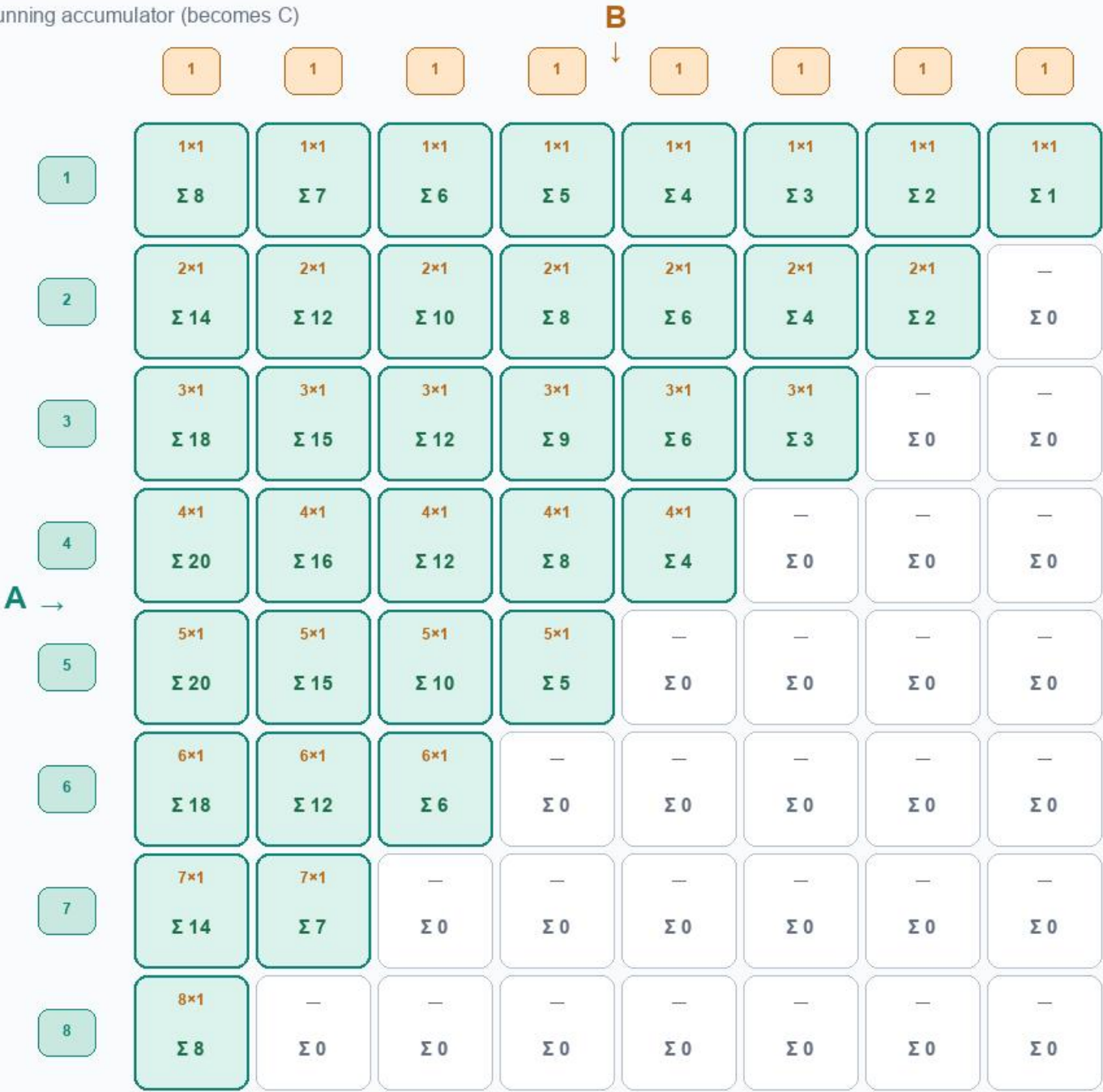
$\Sigma$  = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE



A and B flowing through each PE · Σ is the running accumulator (becomes C)



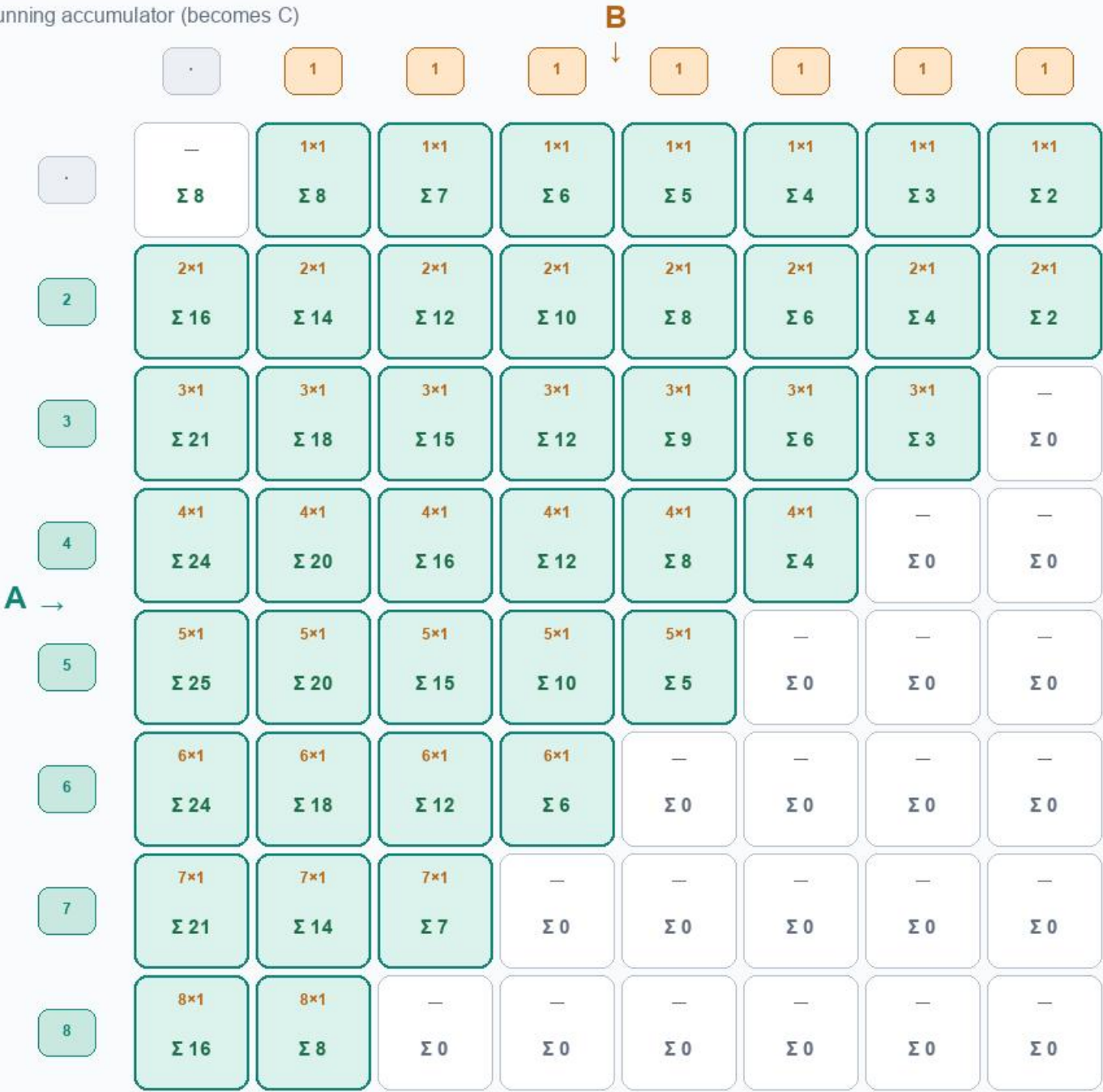
cell: A×B (operands this cycle)

Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell: A×B (operands this cycle)

$\Sigma$  = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell: A×B (operands this cycle)

$\Sigma$  = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

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cell: A×B (operands this cycle)

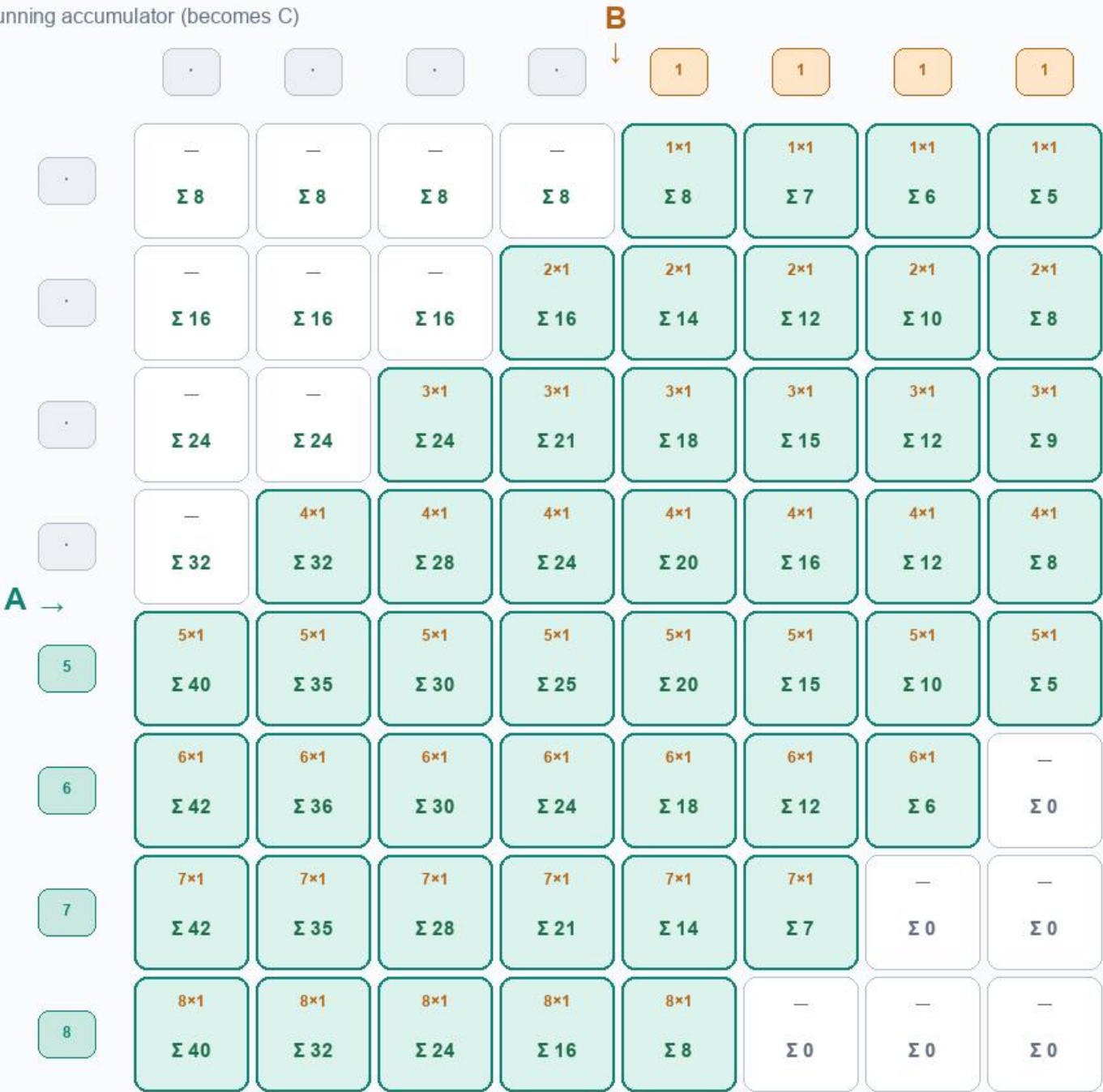
Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE



A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell:  $A \times B$  (operands this cycle)

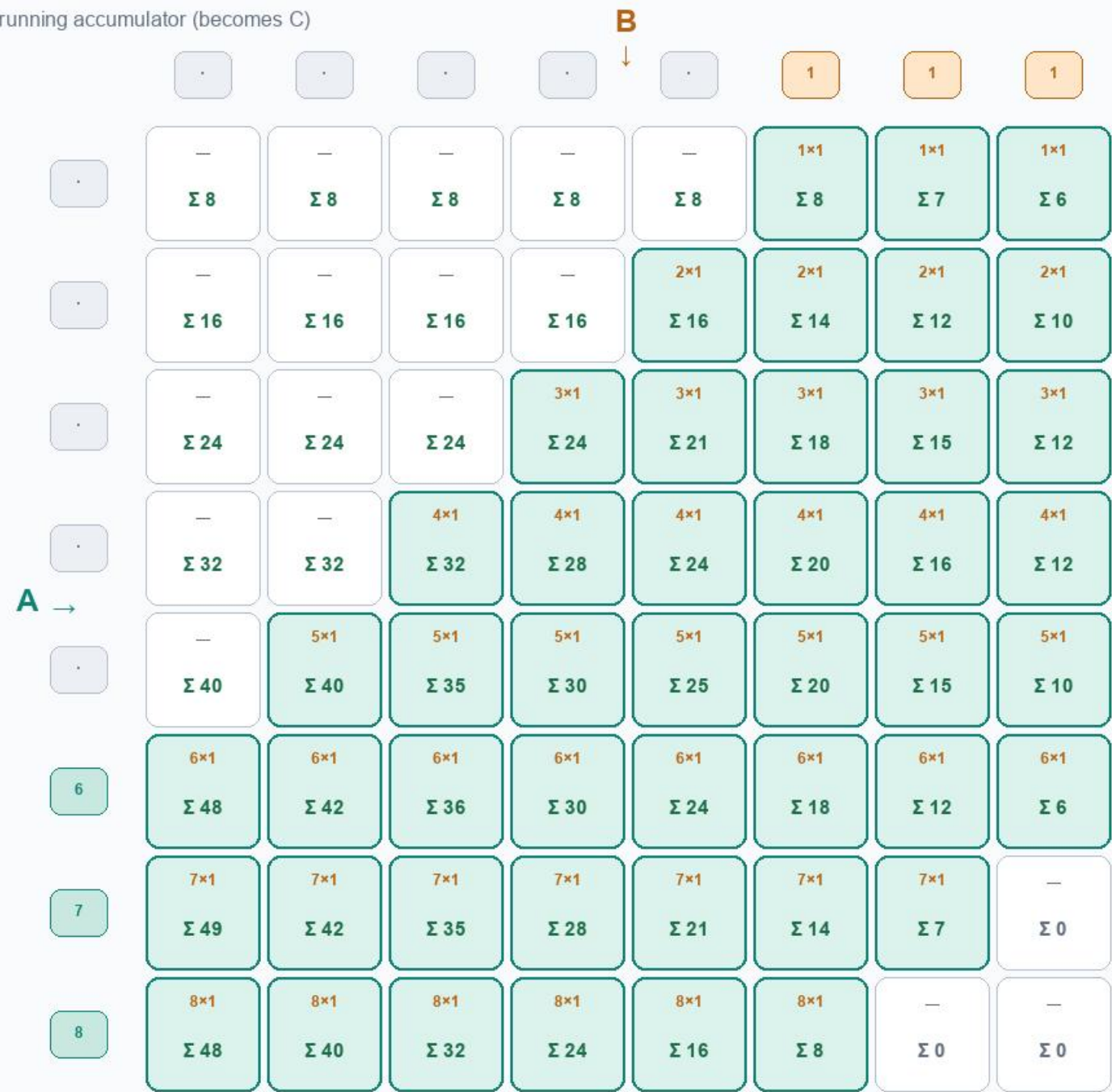
$\Sigma$  = running  $C[i][j]$  accumulator

mint cell = MAC this cycle

C stays in PE



A and B flowing through each PE · Σ is the running accumulator (becomes C)



cell: A×B (operands this cycle)

Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE · Σ is the running accumulator (becomes C)



cell: A×B (operands this cycle)

Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE · Σ is the running accumulator (becomes C)



cell: A×B (operands this cycle)

Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

**Cycle 16 of 22 (t = 15)**

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)

clock 16/22

streaming MACs



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

```
mint cell = MAC this cycle
```

C stays in PE



A and B flowing through each PE · Σ is the running accumulator (becomes C)



A and B flowing through each PE · Σ is the running accumulator (becomes C)



cell: A×B (operands this cycle)

Σ = running C[i][j] accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

mint cell = MAC this cycle

C stays in PE



A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

mint cell = MAC this cycle

C stays in PE

A and B flowing through each PE ·  $\Sigma$  is the running accumulator (becomes C)



cell:  $A \times B$  (operands this cycle)

$\Sigma$  = running  $C[i][j]$  accumulator

mint cell = MAC this cycle

C stays in PE

After 22 clocks each PE holds one  $C[i][j]$ .  $A[i][k]=i+1$ ,  $B=ones \rightarrow C[i][j]=8 \cdot (i+1)$

C

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 8  | 8  | 8  | 8  | 8  | 8  | 8  | 8  |
| 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 |
| 24 | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| 32 | 32 | 32 | 32 | 32 | 32 | 32 | 32 |
| 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 |
| 48 | 48 | 48 | 48 | 48 | 48 | 48 | 48 |
| 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 |
| 64 | 64 | 64 | 64 | 64 | 64 | 64 | 64 |

Result verified against A @ B · matches sim/systolic\_array\_tb.sv